

RISK & FRAUD ANALYTICS

Fraud Detection Analysis Report

Comprehensive Dashboard Findings & Recommendations

1. Executive Summary

We reviewed 6.43 million financial transactions spanning five transaction types. Of these, 8,318 turned out to be fraudulent — a fraud rate of just 0.13%. It sounds small until you look at the dollar figure attached to it: roughly \$12.1 billion moved through fraudulent transactions, which tells us fairly plainly that whoever is behind this isn't bothering with small, everyday payments. They're going after the big transfers.

This edition of the report reflects a few changes made to the dashboard since the last review — most notably, a new metric that tracks how well the system's own automated flagging is actually performing. That turned out to be one of the more important findings in the whole exercise, and we've placed it up front rather than burying it near the end.



Note: the "Average Fraud Amount" card from the earlier dashboard build has been retired in favor of Fraud_Detected, which we think is a more useful number for decision-makers — see Section 2.

2. Understanding the Key Numbers

Total Transactions — 6.43 Million

This is the full dataset under review — 6.43 million individual money movements captured by the system. Every other number in this report is a slice of this total.

Total Fraud Cases — 8,318

Out of 6.43 million transactions, 8,318 were confirmed fraudulent. Put another way, that's about 8 fraud cases for every 10,000 transactions processed — not a lot in relative terms, but each one did real damage.

Fraud Rate — 0.13%

Only 0.13% of transactions were fraudulent. On its own, that's a reasonably encouraging figure — most of the system's traffic is clean. But as the next few numbers show, volume and value tell two very different stories here.

Total Fraud Amount — \$12.1 Billion

Those 8,318 cases add up to roughly \$12.1 billion in losses. Averaged out, that's about \$1.46 million per fraudulent transaction — well above what a typical legitimate transfer looks like. Fraudsters aren't skimming small amounts here; they're deliberately going after high-value transfers, which is exactly the kind of pattern that should shape where controls get tightened first.

Accounts Drained to Zero — 8,112 renamed

This metric was previously labeled "Zero Balance Cases" on the dashboard. We've renamed it because the old label didn't really say what it meant — it's now clearer that this counts transactions where the sender's account was completely emptied out, down to zero, as a result of the fraud. 8,112 out of 8,318 fraud cases fall into this bucket, which

is a striking number. When almost every fraud case ends with an account fully drained, that's the signature of account takeover — someone getting into an account they don't own and pulling everything out before anyone notices.

Fraud_Detected — 16 new

This is a new addition to the dashboard, and arguably the most important number in this report. It counts how many of the 8,318 fraud cases were actually caught in advance by the system's existing automated rule, as opposed to being confirmed after the fact.

The gap between 8,318 and 16 is the headline finding here. When we ran the numbers on the current rule's performance, every single one of those 16 flagged transactions turned out to be genuine fraud — 100% precision, no false alarms. But it only caught 16 out of 8,318 actual cases, which works out to a recall rate of roughly 0.19%. In plain terms: when the system does raise a flag, it's right every time. It just almost never raises one. See Section 9 for what we think should be done about this.

3. Fraud by Transaction Type — Where Does Fraud Happen?

The dashboard breaks fraud down across all five transaction types, and the picture is about as clear-cut as these things get.

Transaction Type	Fraud Cases (approx.)	Risk Level	What This Means
CASH_OUT	~4,200	Very High	Cash withdrawals — the single biggest fraud target
TRANSFER	~4,100	Very High	Account-to-account transfers — nearly as exposed as CASH_OUT
CASH_IN	~0	Low	Deposits — essentially no fraud observed
DEBIT	~0	Low	Debit payments — minimal exposure
PAYMENT	~0	Low	Merchant payments — practically fraud-free

Key finding: CASH_OUT and TRANSFER together account for essentially all fraud in this dataset. The other three types are clean. Both dangerous channels share one thing in common — they move money out of the system quickly and irreversibly, which is precisely what makes them attractive to whoever is running this.

4. Safe vs. Fraudulent — Amount by Fraud

Category	Amount	Share of Total	Interpretation
Safe	\$1.14 Trillion	98.95%	Money that moved without incident
Fraudulent	\$0.0121 Trillion	1.05%	Money involved in fraud (~\$12.1B) reconciled

One housekeeping note: earlier versions of this dashboard showed the fraud amount as roughly "\$10B+" here while the headline KPI card said \$12.09B — close, but not quite matching. That's now been tightened up so both figures agree at \$12.1B. Small thing, but it matters when a client is cross-checking numbers across a page.

1.05% still sounds low at a glance. In absolute dollars, though, that sliver represents over \$12 billion — a reminder that percentages can undersell the real financial stakes when the underlying volume is this large.

5. Who Are the Fraudsters? — Fraud by Customer Type

The dashboard groups fraud accounts into two buckets. This view was previously shown as a donut chart with percentages only; it's now displayed as a line comparing raw counts, which — honestly — makes the size difference between the two groups land a lot harder.

Customer Type	Count	Share	What It Means
One-time	~8.1K	98.72%	Fraudsters who struck once and disappeared
Repeat	~0.1K	1.28%	Fraudsters who came back and struck again

One-time fraudsters dominate

Nearly 99% of fraud accounts only ever appear once. This is the classic "hit-and-run" pattern — an account is created or compromised, used once, and abandoned. It's a hard pattern to catch ahead of time using blacklists alone, since there's no prior history to flag. Behavioral and real-time detection carry a lot more weight here than static rule lists.

Why the small repeat group still matters

1.28% doesn't sound like much, but repeat offenders tend to be the more sophisticated actors — they've found something that works and keep reusing it. Catching and blacklisting this group early has an outsized payoff relative to its size.

6. Where Is the Most Money Moving? — Amount by Type

Transaction Type	Total Volume	Fraud Risk	Action Needed?
TRANSFER	\$0.49 Trillion	Very High	Immediate — top fraud target
CASH_OUT	\$0.40 Trillion	Very High	Immediate — second highest target
CASH_IN	\$0.24 Trillion	Low	Monitor
PAYMENT	\$0.028 Trillion	Very Low	Routine monitoring
DEBIT	\$0.0002 Trillion	Very Low	Routine monitoring

The two highest-volume transaction types happen to be the two most fraud-prone ones. That's not a coincidence worth ignoring — it means the biggest financial exposure and the highest fraud risk sit in exactly the same place, which at least makes prioritization easy.

7. When Does Fraud Happen? — Fraud Count by Day rebuilt

This section looked different in the previous dashboard build. It used to plot fraud counts against raw hourly "steps" (0–744), which is accurate but not something a client can read at a glance. It's since been rebuilt to group those hours into calendar days instead, so the trend line now runs across roughly 31 days rather than 744 individual hour marks.

What the chart tells us now

- Fraud activity starts highest in the earliest days of the period — close to 380–400 cases per day right out of the gate.
- From there it trends downward fairly steadily, settling to somewhere in the 250–280 range by the later days in the window.
- There's some bounce along the way rather than a perfectly smooth decline, but the overall direction is clearly down, not flat.

One thing worth fixing before this goes in front of a client again: the day labels along the bottom of this chart aren't currently sorted in order — they read something like 1, 17, 3, 2, 12, 24, 8... instead of 1, 2, 3, 4. The line itself still plots correctly, but a shuffled axis is the kind of thing that makes a sharp-eyed viewer question the rest of the chart. Worth a five-minute fix (sort the Day field numerically, not as text) before the next version goes out.

Why the decline?

A downward trend over the observation window could mean a few things — fraud rings testing the system early and then backing off once initial attempts succeeded, seasonal or promotional spikes at the very start of the period, or simply the natural front-loading of a synthetic dataset. Without more context on what happened in the business during these specific days, we'd treat this as a pattern worth watching rather than a confirmed cause.

Top 5 High-Risk Hours

Hour No renamed	Total Fraud Cases	Risk Level
6	44	Moderate
212	40	Elevated — active fraud wave
9	34	Moderate
1	32	High — early spike
523	30	Moderate — later period

This column was labeled "Step" in the earlier report. Renamed to "Hour No" here since that's a much clearer description of what the number represents — each unit is one hour of elapsed time.

8. Key Findings Summary

Finding 1 — Fraud is rare, but expensive. Just 0.13% of transactions are fraudulent, yet those 8,318 cases cost \$12.1 billion. The average fraud transaction runs about \$1.46 million, which tells us this isn't opportunistic small-time theft.

Finding 2 — CASH_OUT and TRANSFER are the danger zones. Fraud is concentrated entirely in these two channels. CASH_IN, DEBIT, and PAYMENT show effectively zero fraud, so resources belong where the risk actually is.

Finding 3 — Nearly every fraud case drains the account to zero. 8,112 of the 8,318 confirmed cases left the sender's account at exactly \$0. That's about as clear a signature of account takeover as this kind of data can produce.

Finding 4 — Hit-and-run is the dominant pattern. 98.72% of fraud accounts only ever appear once. Static blacklists won't catch these; the system needs to be looking at behavior, not history.

Finding 5 — Fraud activity front-loads and declines over the period. Rather than a steady rate, fraud counts start high in the earliest days and trend downward across the ~31-day window observed.

Finding 6 — The financial exposure is asymmetric. Safe transactions make up 98.95% of volume, but the remaining 1.05% carries over \$12 billion in losses. Small gains in detection accuracy translate to outsized dollar savings.

Finding 7 — The current automated rule has a serious recall problem. new Of 8,318 confirmed fraud cases, the system's existing flagging rule (Fraud_Detected) only caught 16 of them in advance. Every one of those 16 flags was correct — the rule doesn't cry wolf — but it's missing more than 99.8% of actual fraud. This is arguably the single most actionable finding in this report.

9. Recommendations — What Should Be Done Next?

In rough order of priority:

Priority 1 (Immediate) — Harden CASH_OUT and TRANSFER. Since these two channels account for essentially all fraud, they deserve the strongest controls available: two-factor authentication on large transfers, extra verification for CASH_OUT above a set threshold (say, \$50,000), and a short manual-review delay on first-time large transfers.

Priority 2 (Immediate) — Close the detection gap behind Fraud_Detected. A rule that's right 100% of the time but only fires on 0.19% of actual fraud is, functionally, barely doing its job. This should be treated with the same urgency as Priority 1. Widening the flagging criteria — even at the cost of some false positives — would likely catch far more of the 8,302 cases currently slipping through unflagged. This is a natural next step for whoever owns the detection rule logic.

Priority 3 (Immediate) — Flag zero-balance events in real time. Right now, the "drained to zero" pattern is only visible after the fact. An automatic alert the moment an account hits exactly \$0 would catch many of these 8,112 cases before the money actually leaves.

Priority 4 (Short-term) — Bring in behavioral detection. Since most fraud is one-time, blacklists alone won't cut it. A model that flags unusual behavior — a first-ever large transfer from a dormant account, for instance — would do more here than static rules.

Priority 5 (Short-term) — Fix the day-axis sorting issue on the trend chart. Small, but worth doing before the dashboard goes in front of another client — an unsorted axis undermines confidence in an otherwise solid chart.

Priority 6 (Medium-term) — Build an early-warning system for repeat offenders. Track device IDs, IP addresses, and behavioral fingerprints to catch the small but persistent group of repeat fraudsters before they open a new account under a different name.

Priority 7 (Ongoing) — Review the dashboard on a regular cadence. This is already the second revision of this report in a short window. Fraud patterns move fast; a weekly or biweekly review keeps the numbers — and the controls built on them — current.

10. Conclusion

The overall picture hasn't changed since the first version of this report: fraud here is rare by count but severe by value, almost entirely confined to CASH_OUT and TRANSFER, and dominated by one-time, hit-and-run behavior that drains accounts to zero. What's changed is that we now have a clearer view of how well the existing detection rule is actually performing — and the honest answer is, not very well. It's accurate when it speaks up, it just almost never speaks up.

The good news, as before, is that a concentrated problem allows for a concentrated fix. Tightening controls on two transaction types, adding a real-time zero-balance alert, and — now — expanding the detection rule's coverage would go a long way toward closing the gap between \$12.1 billion in fraud and the 16 cases currently being caught in advance.

A 1% improvement in fraud detection across this dataset would prevent
over \$120 million in losses.

The case for investing in better fraud controls speaks for itself.